GENERATIVE AI: Introduction

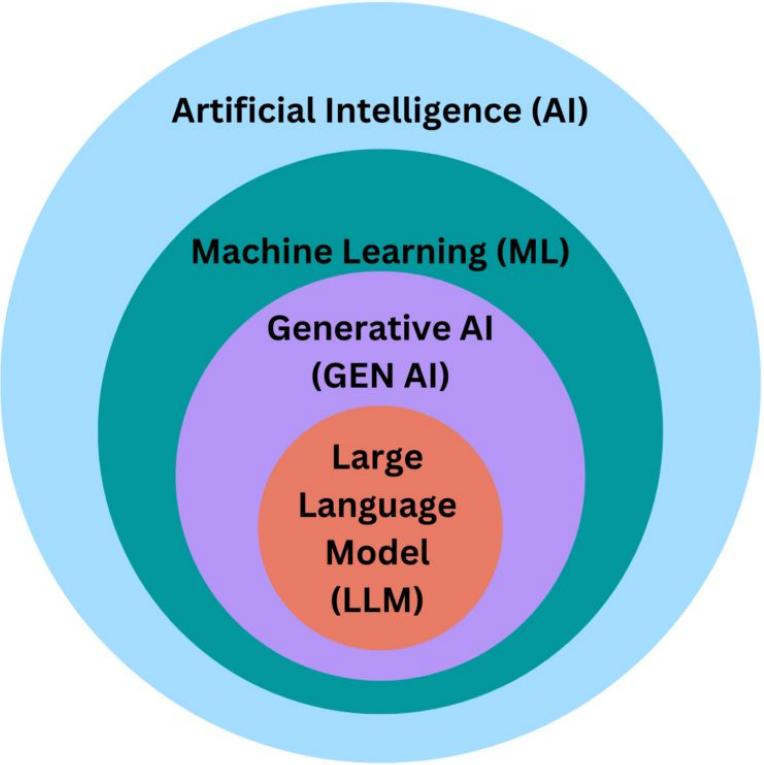
What is Generative AI

Generative AI refers to algorithms that can generate new content, such as images, text, or music, based on training data. It operates on the principles of learning patterns and structures from existing data to create new, similar outputs.

The process of learning from existing content is called training and results in the creation of a statistical model.

When given a prompt gen ai uses this statistical model to predict what an expected outcome might be and with this it gennerates new output.

Where Gen AI Exists ?



Discriminative vs Generative Models

Discriminative models are used to classify data by learning the decision boundary between different classes. They predict labels given features and focus on $P(y|x)$.

Generative models aim to model the joint probability distribution $P(x, y)$. They can generate new data points, making them suitable for applications like image generation.

History of Generative AI

Early Concepts

The idea of machines creating content dates back to the 1950s.

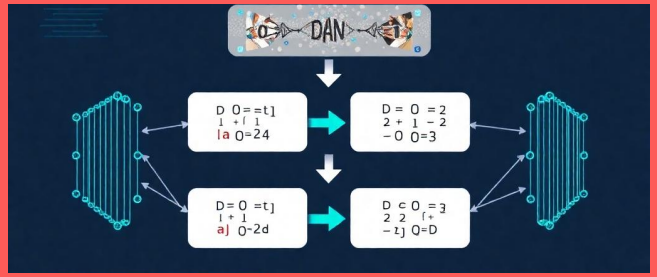
2014

Introduction of **Generative Adversarial Networks (GANs)** by Ian Goodfellow.

2017

Development of **Transformers**, revolutionizing natural language processing.

Understanding GANs (Generative Adversarial Networks)



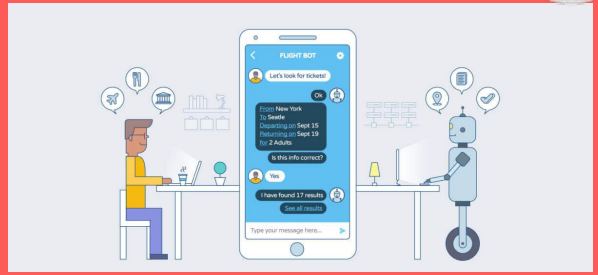
Structure and Function

GANs consist of **two neural networks**: the Generator and the Discriminator. The **adversarial process** involves the Generator creating data while the Discriminator evaluates it, leading to improved outputs over time.

Contribution to Data Generation

GANs are widely used for generating realistic images, enhancing data for training models, and creating art.

Understanding Transformers



Role in Generative AI

Transformers are a type of neural network architecture that excels in processing sequential data, particularly in natural language processing.

Applications

Transformers are used for **text generation** in chatbots and content creation, and for **translation**, enhancing the accuracy of language translation services.

Model Types

Text to text

Text to image

Text to video/3D

Text to task

Foundation Models

A Foundation Model is a large-scale, pre-trained AI model that serves as the base for multiple downstream tasks. These models are trained on massive datasets and can be fine-tuned for specific applications, making them versatile and widely applicable.

Foundation Models

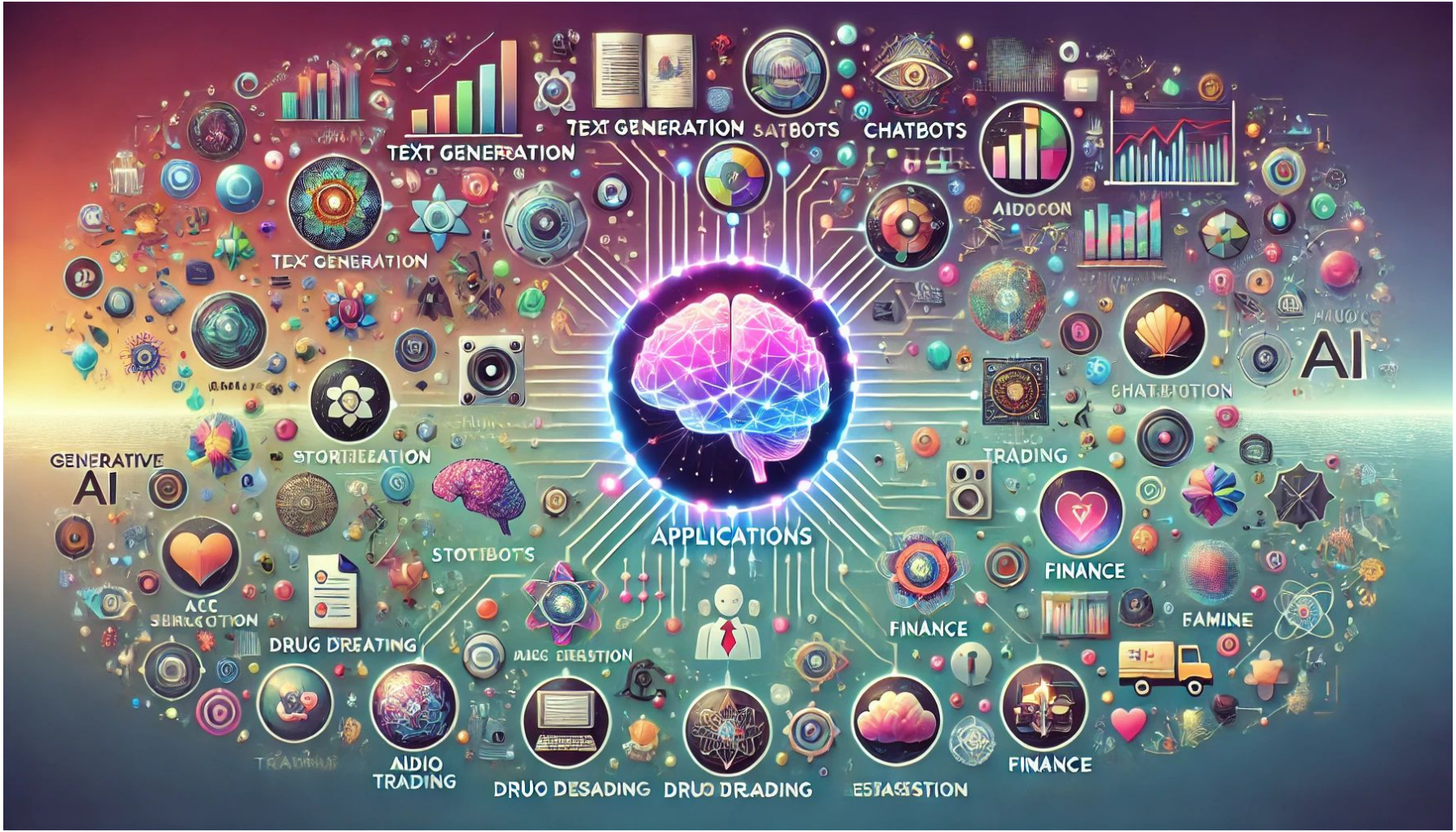
Massive Scale: Trained on billions of parameters and vast datasets.

Versatility: Capable of performing diverse tasks (e.g., language translation, text generation, image recognition).

Pre-trained: Developed through extensive training on general data, requiring minimal additional fine-tuning.

Reusable: Acts as a common foundation for different applications.

Eg. GPT, BERT, T5, DALL-E, Stable Diffusion


TEXT GENERATION
TEXT GENERATION
SATBOTS
CHATBOTS
AIDOCON
TEX GENERATION
AI
CHATBOTION
GENERATIVE
AI
STORTBEATION
TRADING
APPLICATIONS
STOTIBOTS
FINANCE
ACC
SHARGOTION
FINANCE
FAMINE
DRUG DREATING
MICG EIRASTION
AIDIO
TRADING
DRUO DESADING
DRUO DRADING
ESTASSTION
FINANCE

WHY GENAI IS TRANSFORMATIVE FOR SCIENCE?

Reason 1: Accelerates Discovery Speed

Traditional Timeline:

Literature review: 3 months, Hypothesis generation: 2 months, Experimental design: 1 month, Total: 6 months before first experiment

With GenAI:

Literature review + summarization: 1 week, Hypothesis generation (AI suggests based on patterns): 1 week, Experimental design (AI optimizes parameters): 1 week, Total: 3 weeks (5-10X faster)

WHY GENAI IS TRANSFORMATIVE FOR SCIENCE?

Real Case Study - Protein Folding (AlphaFold):

Problem: Scientists needed 50+ years of combined human effort to understand how proteins fold AlphaFold

Solution: AI predicted 200 MILLION protein structures in weeks
Impact: Drug discovery now starts with known protein structure (saves $500M+ per drug)

Time Saved: 5-10 years per drug discovery

WHY GENAI IS TRANSFORMATIVE FOR SCIENCE?

Reason 2: Enables Analysis of Massive Data

Example - Materials Science:

Testing all possible material combinations = infinite

Traditional experiments = one combination at a time

GenAI approach: Generate candidate materials computationally, test top 10

WHY GENAI IS TRANSFORMATIVE FOR SCIENCE?

Reason 3: Democratizes Expertise

Without GenAI:

Healthcare: Only experts in high-income countries have diagnostic tools

Materials Science: Only big companies can afford research infrastructure

Agriculture: Farmers in developing regions can't access crop optimization

With GenAI:

A farmer in rural India can use Gemini (free) to diagnose crop disease from a photo

Researchers anywhere can access AlphaFold to predict protein structures

Healthcare workers can summarize complex medical literature instantly

WHY GENAI IS TRANSFORMATIVE FOR SCIENCE?

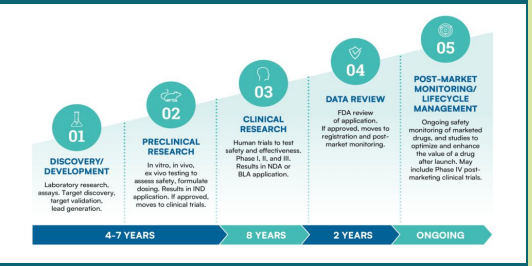
Cost Impact:

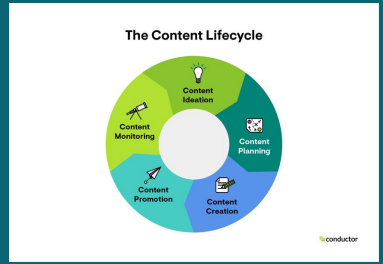
AlphaFold license: $0 (free, open-source)

Traditional protein structure prediction: $500K+ per protein

Impact: 200 million protein predictions instantly available to everyone

Real-World Applications Across Industries





Healthcare

Drug Discovery: Accelerating the process of finding new medications. **Medical Imaging**: Enhancing image quality for better diagnosis.

Entertainment

Content Creation: Generating scripts, music, and artwork. **Video Games**: Creating dynamic environments and characters.

Marketing

Personalization: Tailoring content and advertisements to individual preferences. **Market Analysis**: Generating insights from consumer data.

ETHICAL, REPRODUCIBILITY & BIAS CHALLENGES

Challenge 1: Bias in Training Data

The Problem:

GenAI learns from internet data, which reflects human biases

Historical hiring data → AI learns sexism

Medical data from specific populations → poor performance on others

Real Example - Bias in Hiring:

Amazon built recruiting tool trained on historical hiring data

Tool discriminated against women (because more men hired historically)

Company scrapped the system after discovering bias

ETHICAL, REPRODUCIBILITY & BIAS CHALLENGES

Challenge 2: Hallucinations & False Citations

The Problem: GenAI can confidently generate false information

Might cite research papers that don't exist

Sounds authoritative, but wrong

Real Example: Lawyers used ChatGPT to file legal brief, Citations were fabricated (cases didn't exist), Lawyers fined $5,000 + reputation damage

Why It Happens: GenAI trained to predict "what word comes next" based on patterns, No built-in "fact checker" or access to real-time information, Sometimes pattern completion sounds right but is wrong

ETHICAL, REPRODUCIBILITY & BIAS CHALLENGES

Challenge 3: Reproducibility Issues

The Problem:

GenAI outputs have randomness (same prompt ≠ same output always)

Makes research hard to reproduce

Challenge 4: Intellectual Property & Copyright

The Problem:

GenAI trained on copyrighted works without permission

Generated outputs might resemble copyrighted material

Who owns AI-generated content? Creator or AI company?

ETHICAL, REPRODUCIBILITY & BIAS CHALLENGES

challenge 5: Misinformation & Deepfakes

The Problem:

GenAI can create convincing but false content (text, images, videos)

Can be weaponized for political manipulation, fraud

Real Example:

Deepfake videos of political figures saying things they never said

Potential to manipulate elections, spread disinformation

ETHICAL, REPRODUCIBILITY & BIAS CHALLENGES

Challenge 6: Environmental Impact

The Problem:

Training large GenAI models uses ENORMOUS energy

GPT-3 training: ~1,000 MWh electricity (carbon footprint of 250+ homes for a year)

CHALLENGE 7: DATA PRIVACY

The Problem:

GenAI trained on internet data → might have user PII

Model might accidentally leak sensitive information

BEST PRACTICES FOR RESPONSIBLE GENAI USE

1. Human-in-the-Loop: AI assists humans, doesn't replace decisions
2. Transparency: Disclose when AI is used, what model, known limitations
3. Verification: Double-check facts, especially in high-stakes domains
4. Diversity: Ensure diverse perspectives in AI development + testing
5. Documentation: Record prompts, parameters, results for reproducibility
6. Ethics Review: For high-impact applications, review ethical implications
7. Continuous Monitoring: Track model performance over time, watch for drift

The Future of Generative AI: Trends

Emerging Trends:

- **Advancements in Technology**: Improved algorithms and processing power.
- **New Applications**: Expanding into areas like virtual reality and autonomous systems.

Potential Impact: Generative AI is expected to transform industries by enhancing creativity, efficiency, and personalization.

GENERATIVE AI: For Research and Data Engineering

"Data is hard to get. GenAI generates it for you. GenAI cleans it for you. GenAI analyzes it for you."

The Researcher’s Bottleneck

- The Problem:
  - Real data is expensive, scarce, or privacy-locked (e.g., Patient records).
  - Real data is "messy" (Missing values, wrong formats).
  - Rare events (e.g., Factory explosions, Rare diseases) almost never happen in real datasets.
- The Consequence: Your model fails because it hasn't seen enough "edge cases."

Trinity of Data

1. Real Data: Ground truth. Noisy. Expensive.
2. Simulation Data (Physics-Based):
    - Rules-based (e.g., Finite Element Analysis, Weather models).
    - Good for: Things we understand perfectly.
3. Synthetic Data (GenAI-Based):
    - Probabilistic (LLMs, GANs).
    - Good for: "Hallucinating" realistic scenarios, text, and filling gaps.
    - Key Distinction: Simulation follows Rules. GenAI follows Patterns.

Why GenAI is the "Universal Glue"

- **The Old Way:** Writing 1000 lines of Python code to clean data or using Regex.
- **The GenAI Way:**
  - Data Cleaning: "Hey LLM, fix the typos in this column."
  - Data Augmentation: "Generate 50 more examples of 'Drought' conditions based on these 3 real examples."
  - Multimodal Fusion: Combining Text notes + Sensor numbers + Images.

Case Study 1: Digital Twins for Smart Manufacturing

- A digital twin is a virtual clone of a real-world object, updated continuously with live data so you can monitor, simulate, and improve the real thing.
- A digital twin is a digital model of a physical system that:
  - Collects real-time data from sensors
  - Mirrors the physical system's state (health, performance, environment)
  - Simulates possibilities (what-if analysis)
  - Predicts failures, optimizes performance
  - May even act autonomously using AI

Real-World Uses

- Manufacturing – Predicting machine failures, optimizing production
- Healthcare – Patient digital twins for personalized treatments
- Smart Cities – Traffic optimization, disaster planning
- Automotive – Tesla uses digital twins of each ca
- Aerospace – NASA used it for Apollo missions (before “digital twin” was a term)
- Building & Construction – BIM + digital twin = smarter buildings
- Energy – Wind turbines with real-time performance data

Case Study 2: Clinical Trial Augmentation Saving Lives with Fake Patients

- The Problem: Clinical trials are slow, expensive, and privacy-heavy (GDPR/HIPAA).
- The GenAI Solution: Synthetic Control Arms.
  - Instead of recruiting 100 people for the placebo group, generate 100 "Synthetic Patients" based on historical medical data.
- Key Application:
  - Privacy: Share dataset structures without sharing real names.
  - Imputation: If a patient drops out of a trial on Day 20, GenAI can predict their Day 30 status based on thousands of similar patients.
- Result: Faster drug approvals, zero privacy risks.